

Chapter 2

Defining tonal inventories: Criteria for toneme identification

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1 Introduction

A fundamental question in the typological study of languages concerns what should be used as the units of comparison. Phonological typology seeks to establish cross-linguistic comparison primarily with regard to two types of phenomena: phonological inventories and phonological processes. For a typologist, inventories are usually simpler to identify than processes (Maddieson 2023). Phoneme inventories are almost universally present in language descriptions, have highly conventionalized means of representation, and are conceptually simpler than dynamic processes. Yet paradoxically, for the typology of tone the opposite seems to be the case. We have a solid typological view of tonal alternations, also known as tonal sandhi (Hyman & Schuh 1974, Hyman 2007a). But there is a great deal of confusion and a lack of consensus on how the inventory of a tonal system should be defined.

One commonly finds a discussion of the tonal inventory under the heading “number of tones” (Gordon 2016, Maddieson 2013). It is well established that underlying tonal contrasts do not necessarily map transparently onto surface pitch, as evidenced by tonal alternations. Therefore the widespread use of the term “tone” in the tonological literature to refer to both surface pitch contours and underlying contrasts generates substantial confusion. Given current usage, “number of tones” could mean (i) the number of contrasting pitch levels in a language (high, low, mid, downstepped high, etc.), (ii) the number of possible pitch

gestures (either level or contour) “realized without interruption by a (supralaryngeal) consonant” (Hyman & Leben 2021: 49), or (iii) the number of phonological units with the potential to distinguish lexical and/or grammatical meanings by means of pitch contrast (levels, contours, melodies).

These three ways to count tones can, and often do, give different numbers. For example, in Babanki (Akumbu & Vydrin 2024), 4 contrasting pitch levels are distinguished (H, M, L and xL), but only 2 phonological units (/H/, /L/), the others (M, xL) being produced by regular tonal processes or attested in allotones. In Dom (Gerasimov 2026 [this volume]), there are 2 contrasting pitch levels (H, L) and 3 phonological units (/H/, /HL/, /LH/). In Hausa, three pitch gestures are attested on a syllable (H, L, HL), but phonologically they correspond to a simple /H/ vs. /L/ contrast (Litvinova 2024). In Japanese, there are 3 contrastive pitch levels (H, M, L) and only one phonological unit (/HL/), see Polyakova (2026 [this volume]).

For a typological study, comparison of inventories on any of the three levels named above makes sense. But we would argue that the latter, the level of abstract phonological units (iii), is the most relevant to phonology *per se*, and hence to phonological typology (Hyman 2007b). It is also the one where the most theoretical uncertainty and disagreement lies. And what matters to comparative research is a consistent definition of the phonological units that makes them cross-linguistically comparable in the first place.

In this article, we present an approach to identifying the phonological tonal inventory of a language. We regard the tonal inventory as a set of discrete distinctive units that refer to contrastive tonal patterns in a language. We will use the term *toneme* to refer to these units.

As abstract units, tonemes are not directly observable and must be deduced from surface distributional phenomena. The results of this process of deduction are highly dependent on the phonological framework chosen, which brings its own set of assumptions about the representation of tone and tonal processes. For a typological survey, the situation is complicated by the fact that for most languages grammatical descriptions are the only accessible data, while experimental testing of phonological hypotheses is infeasible. The cross-linguistic comparability of toneme inventories then crucially depends on development and consistent application of a set of diagnostic criteria for toneme identification.

In a typological project, we are hesitant to choose a single one of the existing theoretical representational models for the phonology of tone. As an alternative approach, we devise a theoretical framework that aims to minimize the set of required assumptions about the representation of tone. To this end, we seek concepts that are motivated either by function or surface phenomena. The

core notions of the suggested framework include: the toneme as a minimal distinctive unit, the coordination of tonemes with segments that takes place within prosodic units, and tonal processes understood as alternations in the coordination of tonemes with segments. This framework serves as a conceptual basis to define operational criteria which can be used to infer toneme inventories. To ensure consistency, the criteria only employ the concepts defined within the framework, to avoid relying on excessive theoretical assumptions.

The chapter is organized as follows. Section 2 discusses the concept of the toneme and its coordination with segments. Section 3 presents our definition of a tonal process. All criteria for tonemic status are listed in Section 4, grouped into general criteria, criteria for non-zero phonological status of a tone, and criteria for contour tonemes. Section 5 presents a discussion of our contributions, the criteria we have rejected, the comparison of the notion of toneme with morpho-toneme and underlying tone, and some caveats on using our system of assumptions. Section 6 concludes.

2 Toneme: a definition

We start with the following definition of the toneme:

- (1) The toneme is a minimal pitch-based phonological unit which is employed to distinguish lexical and/or grammatical meanings (in a given language).

The toneme is defined by either a contrastive pitch target (a level tone) or a sequence of pitch targets (a tonal contour). In some languages, tonal contrasts may be phonologized jointly with some non-pitch characteristics, such as phonations (Kirby & Brunelle 2017). We acknowledge this possibility but do not discuss it in detail, as it is irrelevant to the aim of the current work: the definition of the tonal inventory on the basis of pitch contrasts.

Our task of defining cross-linguistically comparable tonal inventories requires operational precision, so the above definition needs additional elaboration. In the remainder of this section we will discuss our assumptions about the nature of the phonological representation of tone in more detail. We start with a short sketch of the history of the term (Section 2.1), then discuss the morphological understanding of tone (Section 2.2). The most contested theoretical question is the phonological nature of the coordination between tonemes and the segmental chain, so we briefly outline our position (Section 2.3). To understand the prosodic restrictions on the coordination of tones, we need a notion of minimal prosodic unit (Section 2.4). We proceed to discuss the nature of phonological tonal targets

(Section 2.5), and the idea of contour tonemes (Section 2.6). A brief discussion of the scalability of tonemes follows (Section 2.8). The section concludes by the definition of tonal span as a surface realization of a toneme (Section 2.9).

2.1 Brief history of the term

Since toneme is a venerable term that has already celebrated its hundredth anniversary, it deserves a short historical overview of meaning and usage. The term was born within structuralist linguistics and has a long tradition of use and reinterpretation. It was first introduced by D. M. Beach (1923: 90): “a toneme is a group of tones within a given language, no one of which can occur in the same position as any other”. This definition is indicative of Beach’s vision of the toneme as a direct analogue of the phoneme (evidently, this analogy is not as direct as it may have seemed at the time). Subsequently the term was extensively used by Kenneth Pike, who characterized the toneme as “a significant pitch unit” (Pike 1948: 4).

Pike’s overly straightforward vision of the toneme as a feature of every syllable in a tonal language could not be sustained for long. The instances of mismatch between toneme and syllable, which seemed to Pike to be simply a minor oddity of some languages, was fairly soon identified as a common occurrence. This led to a reformulation of the definition of a tonal language by Welmers a decade later, as cited below, see Section 2.2. The term “toneme” continued to be used, albeit without a clear definition and with gradually declining popularity. Notably, in his seminal work on autosegmental phonology, Goldsmith (1976) uses “toneme” without defining it, but apparently in reference to elementary tonal segments (level tones such as high and low, but not composite contours, which he prefers to label as “tonal melodies”). In modern linguistic publications, the term “toneme” is used occasionally (e.g. Coupe 2014, Bertet 2021); however, authors are most often content with “underlying tone” or just “tone”. For us, the rationale for reintroducing “toneme” into general circulation is not just to replace “underlying tone” for the sake of smoothness of expression. Our conviction is that the substantiation and precise definition of the notion of toneme as a basic unit of tonology has a high heuristic value, as it helps us make our theoretical assumptions explicit.

2.2 Morphological nature of the toneme

Classifying a prosodic system as tonal depends on the definition of linguistic tone. Here we subscribe to the morphological view of the nature of tone first

suggested by Welmers (1959, 1973: 80): “A tone language is a language in which both pitch phonemes and segmental phonemes enter into the composition of at least some morphemes”. Hyman later adapted this definition to give it its current canonical form, whereby a tonal language is one “...in which an indication of pitch enters into the lexical realization of at least some morphemes” (Hyman 2011b: 231). In other words, in a tonal language, an underlying phonological tonal contrast is not necessarily ascribed to every syllable. This definition includes the so-called pitch-stress or “pitch accent” languages as a subcategory of tonal languages proper (Hyman 2011b). Tonal languages thus defined are quite common: recent estimates informed by wide samples of cross-linguistic data suggest that 42% to 43% of the world’s languages are tonal (Maddieson 2013, Maslinsky et al. 2024).

Following the tradition that separates tonal languages from all others, we limit our use of the term *toneme* to only refer to cases where tonal contrast is employed to convey lexical or grammatical meaning. Various intonational tones, tasked with communicating pragmatic meanings, marking boundaries, and filling pauses, are not treated as tonemes and are not subject to the criteria for toneme identification proposed here.

Information on the morphological associations of tonemes is an important instrument for our analysis. We regard each toneme as a part of the exponent (or the sole exponent) of some morpheme. Moreover, it is appropriate to replace the morpheme in this statement with a more general notion of construction, understood as a form-meaning pair (Haspelmath 2023, Morin 2025). Without loss of generality, we may say that a toneme may be part of the exponent of any item in the mental lexicon (or rather the inventory, see Haspelmath 2024). We will call such a unit an *m-item*, to highlight that our framework does not entail any specific understanding of morphemes and is in principle compatible with a wide range of morphological theories, including usage-based ones.

Whenever we will need to refer to the affiliation of a toneme to a particular *m-item*, we will use a special notation: *m*(/H/; *da*) where the part before the semicolon stands for the phonologically specified tonal component of the exponent, and the segmental component is indicated after the semicolon. The *m-item* unites a segmental and a tonal specification, but either of the two can be zero. A single *m-item* can contain more than one toneme.

2.3 Coordination of tonemes with the segmental chain

Any theoretical account of tone must tackle tonal phenomena that demonstrate the relative independence of tones from segments. Tonemes, as phonological dis-

tinctive units, are building blocks for meaningful linguistic signs. But unlike traditional segments, their surface position with respect to other parts of the exponent is more variable. Evidence for the autonomy of tone comes from (i) tonal processes that allow for the spread or displacement of tones onto different segments and indicate the persistence of tonal contrasts even when segments are deleted, (ii) the existence of morphemes that consist of tone only, and (iii) the general non-isomorphism between segmental morphemes and tones communicating lexical or grammatical information (Hyman 2011b, Kaldhol 2024). What varies is not only which toneme, but *where* it should be found in an utterance.

Identification of tonemes is complicated by the fact that normally every syllable is pronounced with some pitch (F_0), but only some of those pitches may carry lexical or grammatical information. Notably, *either* constant placement of a pitch with respect to some segment *or* contextually dependent variation of the position of a pitch may serve as evidence for tonemic status of the pitch. They both signal that the underlying tonal target communicates information that should not be lost in the context and is thus phonologically specified. In our criteria, we rely both on invariant and variable properties of pitch placement to identify tonemes.

The best approach in order to understand the phonological relationship between tonemes and segments is to take the perspective of a speaker. It is the speaker's task to coordinate tone with other pieces of the utterance given the communicative task and the prosodic conditions. So the phonological representation must reflect that coordinative goal in some way.

Articulatory phonology is a theoretical framework that is built on the idea of deriving phonological representations from the articulatory intentions of a speaker. Phonological information is represented in this framework as a set of temporally coordinated articulatory gestures that serve as invariant phonological units (Browman & Goldstein 1992). Tonal contrasts have recently been included in the framework as pitch gestures on a par with other (segmental) gestures (Gao 2008, Mücke et al. 2012, Karlin & Tilsen 2014, Karlin 2022, Kim & Tilsen 2025). While we do not necessarily subscribe to all of the technicalities of articulatory phonology, we believe it provides a good starting point for devising a typological framework with minimal assumptions about the phonological representation of tone.

The first relevant idea is the isomorphism between articulatory intentions and phonological representations. In practical terms, this means that there is no need to posit more phonological units in the representation than there are articulatory gestures in the utterance. A toneme may then correspond to one gesture, or a sequence of gestures. But it does not make much sense to postulate a sequence of different tonemes when the realization can be argued to represent a single

pitch gesture. For instance, a low tone realized across several syllables may still constitute a single phonological unit.

The second idea starts from the basic premise that phonological information must be communicated to the hearer in some way. Then for highly contextually variable phenomena, such as pitch, the phonologically relevant information must be invariant, that is, preserved across contexts. The relative timing of the realization of pitch gestures and segmental gestures is the informative dimension for the question of how tonemes are coordinated with the segmental chain. Whenever timing of the start of the pitch gesture or its target relative to other segmental gestures is invariant, this indicates that it is controlled phonologically. The notion of segmental anchoring (Arvaniti et al. 1998, Ladd et al. 2000), i.e. stable alignment of tones to the segmental string, extensively used to identify underlying tonal targets (Arvaniti et al. 2024), is based on the same rationale. Our notion of coordination of tonemes with segments is a generic term for the same idea.

There are experimental data showing that even small differences in the timing of pitch may serve as perceptual cues to the hearer. This works for both tonal contrasts, as in Serbian (Karlin 2022), and intonational contrasts, as in Greek (Arvaniti et al. 2006). But theoretical interpretations as to what exactly the timing of pitch gestures communicates about the phonology of tone coordination differ significantly among authors.

Authors who embraced the autosegmental interpretation of tone usually proceed from the postulate that coordination of pitch gestures with segments is mediated through such prosodic units as moras or syllables. The phonologist's task is then to attribute a tonal target to the prosodic constituent it is associated with; but beyond that, not much phonological value is seen in the tone timing data. Researchers may admit that differences in pitch timing can communicate information about secondary association, but would rather do without complications to the basic association model unless they are strictly necessary (Arvaniti et al. 2006).

Authors working within the framework of articulatory phonology start from the assumption that pitch gestures are coordinated with other segmental gestures directly. Empirical data on invariance in the timing of tones may help these researchers adjudicate between specific phonological hypotheses about coordination of tones with segments. For instance, studies on Thai and Tibetan present an argument that the relative timing of consonantal and tonal gestures indicates that tones are coordinated in a way similar to additional consonants in a syllable (Karlin & Tilsen 2014, Hu 2016).

In a typological survey, we face a two-sided problem with the analysis of coordination between tones and the segmental chain. For one, our sources for most languages will only include grammatical descriptions that do not necessarily provide enough phonetic detail about pitch timing, let alone perceptual experimental data. But at the same time, we do not want to make strong theoretical assumptions either, to avoid limiting *a priori* the possible ways of coordination between tones and the segmental string.

Our choice here is to limit information about the coordination of tonemes and segments in m-items – our representational level – to the bare minimum required for adequate description. Let us suppose that words in a language have lexical tones, and the distribution of pitch in the wordform can be predicted on the basis of lexical tone. Then the m-item should be a simple pair of a toneme and all segments, with no further specification. For example, in Bambara a toneme is typically mapped onto the whole word (2). So the information in the m-item boils down to the statement that the lexical tone for ‘road’ is High.

- (2) Bambara
m(/H/; sira) → sirá ‘road’

If, however, the distribution of pitch cannot be predicted given the lexical tone, that information should be reflected in the m-item. In these cases we employ the *alignment point* – a special mark that denotes the coordination of the pitch gesture with either a segment or the prosodic constituent edge. We mark the alignment point with a ^ symbol. The ^ in the toneme specification indicates which tonal target is aligned. E.g., for the falling (HL) toneme in Japanese (3) either the right edge of the H target is aligned with the rhyme of the *ko* syllable or, equivalently, the left edge of the L target is aligned with the onset of the *ro*.

- (3) Japanese
m(/H^L/; koko^ro) → kòkòrò ‘heart’

In both examples above, irrespective of the presence of the alignment point, the information contained in the m-item does not imply a specific form of coordinative relationship between tonemes and segments. Whether the pitch gesture is coordinated with a particular segment or a prosodic unit is left unspecified. So different theoretical perspectives on the phonology of tone coordination, including articulatory phonology and autosegmental-metrical models, remain compatible with this approach. We argue that this kind of minimalistic representation is best suited for typological analysis.

2.4 Minimal prosodic unit

To properly discuss the surface realization of tonal targets, we should recognize that possible patterns of coordination between tonemes and segments are limited by articulatory and perceptual constraints. To be perceived, an F_0 pitch gesture must overlap with a vowel or a sonorous consonant. This constraint gave rise to the idea of phonological association of tones to prosodic units.

Association is a holistic relation between a tonal target and a prosodic constituent called its *tone bearing unit* (hereafter, TBU). Association is a key notion for the autosegmental paradigm in phonology (Hyman & Leben 2021). In autosegmental literature, association was essentially defined from the acoustic perspective, as a temporal integration, or overlap, of the tone realization with a syllable rhyme or a mora (Goldsmith 1976, Sagey 1988). This acoustic relationship was then carried over into the idea of the underlying association of tones to TBUs.

Recent research raises doubts that acoustic overlap with the TBU is the optimal way to represent coordination of pitch gestures with segments. Accumulating phonetic data on relative timing of tones indicates that landmarks for tone coordination are defined in the articulatory rather than the acoustic modality (d'Imperio et al. 2007, Burroni & Tilsen 2025). It is unlikely that target simultaneity is the default mode of coordination between tone and segments. For some languages, there is evidence for stable timing of tonal gestures relative to the syllable onset, not the target (Hu 2016, Lundmark et al. 2021, Kang & Xu 2024, Burroni & Tilsen 2025). Late realization of tonal targets is also commonly attested, causing the so-called spillover of tone into the next syllable (Myers 1999, Xu 2001, Burroni 2023). In short, cases of acoustic mismatch between a pitch gesture and its supposed TBU are common. For our task of identifying tonemes from distributional data, discretizing tonal realizations into a sequence of acoustic TBUs may result in undesirable misattributions.

Moreover, the autosegmental view of the tone–TBU association usually comes packaged with a few additional assumptions which we are unwilling to make. There is a longstanding idea that tones are bound to a single level of the prosodic hierarchy, so that constituents of that level serve as TBU. In the current literature, syllables and moras are supposed to be valid TBUs (Yip 2002: 74, Hyman & Leben 2021). This implies that the choice of TBU is a generic parameter of the grammar of a language, reflected in statements like “The syllable/mora is the TBU in this language”. In fact, however, there is ample evidence, including in the chapters of this volume, that syllables and moras may both be relevant to tonal distribution in a given language. Choosing one over the other leads to the necessity of positing additional constraints or rules to work around this apparent irregularity.

There are also many languages where higher levels in the prosodic hierarchy (e.g. feet or words) are clearly relevant for the distribution of tones. The possibility of having these as TBUs has been discussed, but is often dismissed without explicit argument. The notion of tonal domain (Hyman & Leben 2021) was offered in response to account for these phenomena in the autosegmental paradigm. All of these complications seem to misplace the explanation for the observed distributional complexity into additional representational abstractions and superficial derivations. If one does not assume that tones should always link phonologically to the same level of the prosodic hierarchy, the weight of explanation for tonal distribution may be shifted onto lexical information and particular constructions. The latter position is in line with the general findings of construction grammar, and seems to us a promising direction for exploration in tonological typology.

To clearly indicate where our framework deviates from the established tradition, we will abstain from referring to the term TBU in our criteria. But to properly account for the coordination of tone we still need some means of reference to language-specific prosodic characteristics. In order to achieve this, we introduce the notion of the minimal prosodic unit (hereafter, MPU).

The relevance of the MPU can be understood from the articulatory perspective. In articulatory (motoric) terms, prosodic units can be interpreted as groups of coordinated gestures, termed co-selection sets by Tilsen (2014). The MPU reflects the idea that tonal gestures always have to be coordinated with some sonorous segmental articulations in order to be audible. Hence,

- (4) The MPU is a minimal group of coordinated articulatory gestures (co-selection set) that includes a pitch gesture. It is either the syllable or the mora.

The MPU is thus indicative of the distributional and prosodic behavior of tonemes in a language. By default, the syllable is considered to be the MPU in a language, unless there is strong evidence to the contrary. Stating that the mora is the MPU is equivalent to saying that the mora is relevant to the distribution of tones. Data available in most grammatical descriptions can support inferences about articulatory intentions only indirectly. In most cases, one has to resort to distributional phenomena to support the (ir)relevance of the mora to tones in a particular language.

The most generally accepted criterion for relevance of the mora is the asymmetry of the distribution of tonal contours with respect to syllable weight. If tonal contours are only attested on heavy (bimoraic) syllables, this may serve as evidence for coordination of tones with moras. But this standard approach comes

with a caveat. The minimal time required to achieve pitch change is an articulatory bottleneck (Xu & Sun 2002, Xu 2004) that may exert pressure on the duration of a vowel coordinated with it. Indeed, it is well attested cross-linguistically that contour tones strongly correlate with longer vowels and heavier syllables (Gordon 2001). So one has to be careful about positing a syllable weight contrast based solely on the tonal distribution, to avoid circularity.

To be sure that the mora is an appropriate prosodic category that regulates the realization of pitch gestures, we must confirm that it is prosodic structure that limits or transforms tonal articulations in some way, and not vice versa. The influence of moraic structure may manifest itself in the alignment of the tonal spans or tonal melodies, in the choice of the variants of tonal morphemes, etc. So far, we have formulated the following diagnostic criteria that can support the relevance of the mora as MPU:

1. If there is a tonal process where *the number of realized tonal targets* of a morpheme (usually a tonal morpheme) depends on the number of morae in the segmental chain, mora is the MPU.

E.g., in Bunoge (Dogon) the tonal marker of the head noun in an NP is L if the word contains 1 or 2 morae (CV, CV:, CVCV) or LHL if the word contains 3 or 4 morae (CV:CV, CVCVCV, CVCVCV), as shown in (5), see in more detail Vydrin & Heath (2026 [this volume]).

- (5) a. á kèlè
 2PL horn\HEAD
 ‘your (pl.) horn’
 b. á kùnjúgà
 2PL knee\HEAD
 ‘your (pl.) knee’

2. If *the position of a tonal target* is defined taking into account the number of morae in the segmental chain, the mora is MPU.

E.g., in Japanese, the realization of the /HL/ toneme always starts from the second mora of a word: jò.ó.ró.p.pà ‘Europe’ (Polyakova 2026 [this volume]).

In contrast to the TBU, the identification of the mora as MPU does not entail that morae should be relevant for tone coordination in all contexts. In other words, if the mora is MPU, this does not mean that other units of the prosodic

hierarchy (in particular the syllable) are not relevant to the tonal system. The MPU is a purely paradigmatic notion. Examples of tonal systems where both syllables and morae are relevant for tonal distribution are present in this volume (Hausa, Iquito, Japanese, Bunoge).

2.5 Tonal levels

A great deal of theoretical discussion in tonology revolves around the phonological representation of tonal contrasts. From the perceptual and acoustic perspective, pitch level (F_0 frequency) contrast is the basic building block for tone. Languages contrasting two to five distinct levels are attested.¹ Hereafter we will refer to these as *tonal levels*. Tonal levels are not defined in terms of an absolute F_0 frequency range, but only relative to one another (Maddieson 1978). From the articulatory perspective, tonal contrasts are realized via laryngeal articulatory gestures that approximate pitch targets corresponding to tonal levels (Karlin & Tilsen 2014, Hu 2016, Karlin 2018). In a recent Gesture-Field-Register model (Kim & Tilsen 2025), articulatory tonal targets are supposed to be invariant (stored in long-term memory), while surface F_0 variability is derived from the tonal context and the register state. The GFR model corresponds well to a more abstract phonological representation of tonemes as discrete tonal levels.

Tonal levels can be arranged linearly from highest to lowest. But this one-dimensional picture is complicated by voice register operations whereby the whole pitch range shifts down or up, called downstep and upstep respectively. In many tonal languages, register shift operations can convey phonological information. Similarly to the logic of the GFR model, we do not treat downstepped (upstepped) tonal targets as separate tonal levels in a language, as they are defined in relation to the local tonal context (the register range). Yet in some languages (e.g. Baga Pukur, see Rochant (2023)), the tone register operation (downstep) may function as a separate phonological item, which we accept as a toneme, /^u/.

There is a long-standing debate on whether tonal levels should be represented phonologically as a set of elementary units (High, Low, Mid, etc.) or in terms of binary features ([+high, -low], [-high, +low], [-high, -low]). The latter would unify the treatment of segments and tone in the SPE-style phonological framework, but the former is actually the conventional representation, overwhelmingly dominant in descriptive tonological work. We side with the argument that while a feature-based representation may be useful in particular languages, in

¹6-level tonal systems are sometimes mentioned, but we have not yet analyzed data from any such languages.

a wide typological perspective it creates as many conceptual problems as it is supposed to solve (Clements et al. 2011, Hyman 2011a). Hence we adhere to the dominant practice in descriptive tonology of representing contrasting tonal levels as elementary units.

To summarize, we treat underlying pitch targets as phonological primitives that tonemes are built upon. A simple level toneme, for instance, High, then consists of one underlying pitch target at a High tonal level, hereafter written as /H/.

2.6 Contour tonemes

Many tonal languages, especially in Asia, are traditionally described as making use of contrasting tonal contours — pitch gestures that consist of one or two transitions between tonal levels. One topic of theoretical discussion is whether contour tonemes are possible as phonological units or whether they should always be reinterpreted as sequences of level tonemes.

The idea that a tonal contour can have phonological status, especially if it is attested on a short vowel, originally seemed quite natural. However, as language data accumulated it became evident that in many cases a contour on a syllable can be a result of a tonal process combining multiple phonological units. With the advent of autosegmental phonology, much progress in understanding tonal phenomena was gained by representing tone in terms of level primitives (H, L, etc.). This gave rise to the currently mainstream phonological view of tonal contours as a sequence of levels. In support of this position, there is experimental evidence that the relative alignment of tonal targets that form a contour is independent of speech rate, so that contours are better represented as a sequence of articulatory targets corresponding to tonal levels. This is treated explicitly by Ladd et al. (1999) who argue that if a contour were specified phonologically as a single dynamic target (R, for rising tone), the steepness of tone rise would be insensitive to the speech rate. (For an alternative view on contour tones as dynamic articulatory targets in Mandarin, see Kang & Xu 2024.)

Still, from a more functional perspective, the idea remains appealing that in some languages the carrier of distinctive function is in fact the tonal contour as a whole. A compromise position favoured by autosegmental phonology takes levels as phonological primitives, but any tonal contour may be referred to as a “melody”. What remains unclear is whether tonal melodies are included as a proper part of the inventory of tonal contrasts, or whether they are used just as convenience tools to describe higher-level phenomena (tonotactics, lexical

distribution of tone, etc.). If we accept contrasting tonal contours we should admit contour tonemes into our tonal inventory.

The theoretical basis for positing contour tonemes as structurally complex entities (phonologized sequences of tonal targets) can be found in usage-based linguistics. It is well established that items which are frequently used together and highly mutually predictable show signs of automaticity and holistic processing. The cognitive process that leads to this result is called entrenchment (Langacker 1987: 59, Blumenthal-Dramé 2012). From the perspectives of both speaker and listener, entrenchment serves the goals of effort minimization and compression of information. Importantly, entrenchment is a gradual phenomenon, so that a decision to include a tonal contour in the phonological inventory (as a toneme) implies some arbitrary threshold.

A parallel with segmental phonology can support contour tonemes, since there is ample evidence for articulatorily complex segments (combining multiple non-simultaneous articulatory gestures) that function as phonological units (affricates, prenasalized consonants, palatalized consonants, true diphthongs). A distinct phonological nature of complex segments that makes it possible to distinguish them from phonetically similar sequences of segments can be confirmed either with distributional data on component cooccurrences (Gouskova & Stanton 2021) or with experimental data on the relative timing of articulatory gestures (Shaw et al. 2021). It is then conceivable for a combination of pitch gestures to also be phonologized as a contour toneme.

The problem with evaluating tonal contour entrenchment in a typological project is that for most of the surveyed languages the data will be limited to a grammatical description, probably supplemented with only a short text. That is, neither psycholinguistic nor corpus data is available to empirically test the degree of entrenchment of a particular tonal contour. Hence the need to define heuristic criteria for distinguishing between phonologically atomic and composite contours. The evidence for these criteria should be of the kind that is obtainable from a grammatical description: attested contextual alternations, distributional patterns, lexical items.

There are two possible starting points for classifying tonal units of a language into level or contour tonemes. One approach takes the contour interpretation to be the default, unless there is evidence to the contrary. The idea that a contour is a unit *unless* it is produced by a tonal process appeared rather early (Maddieson 1978: 337). What makes it operationally problematic is the lack of a principled answer to the question of what scale it should be applied at: syllable, morpheme, word, utterance? Data from different tonal languages may arguably support any

of these scales. Uncertainty in this respect would be a huge impediment to comparability of the tonal inventories.

The opposite stance is to interpret tonal contours as a sequence of levels by default, unless there is evidence to the contrary. This position is most clearly articulated by Yip:

for the theoretical linguist committed to universal grammar, if contours in one language are sequences they are presumably sequences in all languages, in the absence of evidence to the contrary (Yip 2002: 50).

While we would distance ourselves from her universalist approach, we agree with Yip that the tonemic character of a tonal contour in each language needs to be supported by evidence. And this is precisely what our criteria (Section 4.3) are designed to achieve.

Yip discusses phenomena that may signal the phonological unity of a contour at some length (2002: 47–52). She enlists two critical sources of evidence. One refers to the structure of the contrastive inventory and essentially boils down to Trubetzkoy’s idea of inseparability (Trubetzkoy 1969), in the same way as it is applied to complex segments (Gouskova & Stanton 2021). We see this as valid and use it in our criteria (Section 4.3.1). The other source of evidence is provided by phonological processes in which a contour participates as a whole unit rather than its component parts. For the latter source, we do not accept her arguments in full (see Section 5.1.4 for details).

In summary, we admit both level and contour tonemes, with the reservation that the phonological unity of a contour should be supported by at least one of our criteria. We define the internal structure of contour tonemes as a sequence of tonal targets (tonal levels). For instance, a rising toneme in a language with two contrasting tonal levels consists of two targets, a Low followed by a High, and is written as /LH/. Whenever we mention a tonal melody, it refers to a stable combination of tonemes that is not itself part of the inventory of phonological tonal contrasts.

2.7 Allotones

Some languages provide clear evidence for the complementary distribution of certain pitch gestures, which allows us to claim they are phonologically equivalent. Following a structuralist terminological model we can call these complementary realizations *allotones* of the same toneme.

We consider as allotones those variants which differ by the set of phonologically specified tonal targets. We do not consider as “true” allotones (and refer

the them as just “variants”): intonationally conditioned variants of toneme realizations; variants which differ by a pitch level conditioned by its position in an utterance, such as automatically downstepped realizations; variants conditioned by transition to the adjacent tonal targets; variants conditioned by transition from the adjacent consonants (tone-depressors or anti-depressors) which do not alter tonal targets.

One of the allotones of a toneme can stand out among the others by its frequency in texts and/or by the number of contexts where it appears. We can refer to it as the main allotone (the other allotones being secondary). However, a toneme may have no main allotone, as in Bambara where both allotones of the toneme /L/, [L] and [LH], are equally frequent in texts.

Our notion of allotone is clearly demonstrated by languages where the contour or level character of a toneme is not absolute. For example, in Lhasa Tibetan (Perekhvalskaya 2026 [this volume]) there are two tonemes (let us refer to these as “toneme 1” and “toneme 2”), and each has two main allotones. Toneme 1 is realized as a falling pitch on a light syllable, and as a level H on a heavy syllable or on a disyllabic foot. Toneme 2 is realized as a level L on a light syllable, and as a rising contour on a heavy syllable or on a disyllabic foot. The tonemes can be regarded as contour (falling and rising) or level (high or low) depending on which allotones we decide to consider as the main ones.

In Bambara (Vydrin 2026a [this volume]) there are again two tonemes. One is always realized as high, and the other one has two allotones: LH if followed by another L, and L otherwise. The latter toneme can be regarded either as a contour (LH) or as level (L), as both allotones are equally frequent.

An alternative interpretation of the phenomena in Lhasa Tibetan and Bambara is to postulate a phonological tonal process conditioned by the tonal context. The problem is that the process can be equally well described as either creation or deletion of a tonal target. To choose one over the other, additional assumptions about underlying representation are necessary. The interpretation of variant realizations as allotones has the advantage that the two tonal realizations are symmetrical, so one does not have to assume that one of the forms is primary and the other derived. This symmetry, of course, comes at the cost of the additional theoretical construct of an allotone.

2.8 Scalability of tonemes

What distinguishes tonemes from other typical phonological segments is a property that we may call scalability. Due to the articulatory nature of pitch gestures

they can be relatively easily sustained for prolonged periods of time, spanning more than one minimal prosodic unit.

The scalability property is motivated by the extensive evidence found across the tonal languages of the world that both level tones and tonal melodies demonstrate a capacity to expand onto neighboring syllables and words. There is an especially pronounced tendency for tones to expand to the right (Hyman 2007a). This tendency agrees with the phonetic data showing that tonal targets tend to be realized late (Gordon 2016: 236). Thus it is important to note that a sequence of similarly-toned MPUs may be a realization of a single pitch gesture, and hence an instance of a single extended toneme.

The flip side of extensibility of a toneme is compressibility. If the duration of a toneme can be varied without compromising its phonological identity, naturally it can be shortened as well as lengthened. There are still articulatory constraints on the minimal time required to attain a pitch target, as discussed in Section 2.4, which limit the compressibility of tonemes.

2.9 Tonal span

If the underlying phonological relationship between tonemes and segments is best understood from the speaker's perspective, the hearer's perspective is a better fit for discussing surface realization of a toneme. Since in many languages tones may spread across several syllables or words, and we do not make the assumption that every syllable/mora has to carry its own toneme, the operational question here is to detect the boundaries of a surface realization of a toneme.

To this end, we introduce the concept of *tonal span* – the part of the segmental chain on which a toneme is realized on the surface level. Ideally, a tonal span should be seen as a perceptual category, in the sense that it encompasses all of the prosodic constituents that are perceived by a native speaker as carrying one and the same toneme. In practical terms, the required experimental perceptive data will be missing for most if not all languages in a typological survey. So we see tonal transcriptions presented in descriptive grammars as a basis for approximating the perceptual tonal spans, as linguists presumably rely on the intuition of native speakers along with acoustic data while doing transcriptions.

To indicate the boundaries of the tonal spans of tonemes we will use square brackets, as shown in (6).

- (6) a. Hausa
màgàrúbà: [L mà.gà][H .rú][L .bà,à] ‘sunset’

b. Lhasa Tibetan

[H tsám.tsám]-là sometimes-in ‘sometimes’

Each bracketed zone corresponds to a single toneme. The toneme is specified right after the opening bracket as a sequence of letters indicating tonal targets. The text inside the brackets corresponds to the part of the segmental chain where the toneme is realized. We assume that a realization of a toneme cannot be interrupted by another toneme, and thus tonal spans cannot be nested. We do not include intonational (boundary, default) tones in tonal spans as we do not regard them as tonemes, following our morphological definition of the toneme.

In most cases, the coordination of the pitch gestures constituting a toneme with the relevant segments can be represented quite naturally by including the corresponding segmental chain (a syllable, a word, etc.) within the brackets of a tonal span. For level tonemes the brackets of a tonal span will correspond to the left and right edges of the pitch gesture. But for contour tonemes, the opening and closing brackets of a tonal span may be insufficient to describe the distribution of the pitch gestures over the tonal span. Alignment points (Section 2.3) can be employed to specify the pitch placement.

In general, a surface pitch level target realized on a single syllable may be the exponent of a toneme, or a part of its exponent, or it may represent more than one toneme, or none. One special case, quite common in African languages, is when a floating low tone is realized as downstep of the following high tone. This realization merges a phonologically relevant register operation (downstep, a lowering of the tonal “ceiling” for the rest of the phrase) with the high pitch target of the following toneme. For notational convenience, we posit a zero-width tonal span for the low toneme at the beginning of the constituent of the following high toneme (7).

(7) Bambara

jíri	‘bée
[H jí.rí]	[L ¹][H bée]
tree\ART-PL all	
‘all the trees’	

3 Tonal process: a definition

We define a *tonal process* as a situation where surface realizations of the same m-item in different contexts are not phonologically equivalent with respect to

tone. This refers to alternations where either a set of tonal targets, or their position relative to other segments of an m-item varies systematically depending on the context.

Alternations that match our definition of a tonal process may arise in three distinct ways.

- (i) Whenever a tonal target is realized outside of the segments of an m-item to which it belongs. That is, a tonal process is attested when a tonal span crosses morpheme boundaries. In this situation the other m-item with which the tonal target overlaps will necessarily alternate with respect to tone. Tonal spread and tonal shift are prototypical examples of this class of processes. All purely tonal morphemes (with no segmental specification) also necessarily produce tonal processes of this kind.
- (ii) In other cases the tonal target is shifted relative to the segments of its morpheme, or realized only in a specific tonal or morphosyntactic context. Spread of a neighboring toneme resulting in the displacement of a tonal target may serve as one example; another is the selection of a different allotone in a specific tonal context. The difference from the previous class is that the alternating tonal target does not cross the segmental boundaries of the morpheme to which it belongs.
- (iii) Tonal processes may be a result of the interaction of lexical or grammatical tones with intonational tones. Intonational tones may affect the position or the fact of the realization of lexical and grammatical tones, and vice versa. Processes of this kind are often attested at prosodic boundaries, e.g. the end of the intonational phrase.

Tonal processes complicate the phonological interpretation, since a tonal target realized on some segment belonging to a particular m-item may in fact be projected from another m-item. Sometimes a complex distributional analysis is required to isolate the tonal specification of each particular m-item in the context. But whenever this is accomplished, this opens up the possibility of applying the diagnostic criteria described in Section 4.

4 Criteria for toneme identification

We approach the problem of defining a language's tonal inventory with no *a priori* assumptions of phonological status for any observed pitch gestures, either level or contour. Phonological status (or tonemic status, as we shall call it) is

only assigned when a tone meets one or more of the criteria suggested in this paper. Meeting the conditions of more criteria strengthens the argument for the tonemic status of a given tone, but any single criterion is logically sufficient.

The idea of defining a list of criteria for identifying the phonological status of tones is not exactly new. Our major predecessors here are Maddieson (1978), Hyman (2000) and Yip (2002). The need to define our own list of criteria is due to the different set of theoretical assumptions about the phonology of tone employed in this typological project. To ensure consistency, the following criteria should be based on our understanding of the notions of toneme (Section 2) and tonal process (Section 3). We have strived to make the criteria clear and operational, providing a full rationale together with examples when necessary.

The idealized procedure for toneme identification includes the following analytical steps:

1. Establish the presence of a phonological tonal target in a certain context to distinguish it from the lack of a tonal target, i.e. zero (see Section 4.2.1). As any syllable is realized with some pitch, one needs to distinguish between the presence of a tonal target and pitch interpolations between targets, realization of a single target over several syllables, default tone, register operations or intonational phenomena.
2. Identify the linguistic function of the target. As we only treat as tonemes those tones that convey lexicogrammatical meanings, this means weeding out tonal gestures that serve intonational functions, as well as default tones.
3. Group certain sequences of tonal targets into contour tonemes, where appropriate.

On a cautionary note, granting tonemic status to a level tone or a contour does not automatically mean that this level/contour will be a toneme in all contexts. Our system does not logically preclude the same pitch gesture from having different phonological interpretations (including lexical vs. intonational, zero vs. non-zero) in different contexts.

In order to identify the tonal inventory of a particular language two major prerequisites should be met: we assume that (i) pitch ranges are already classified into distinct tonal levels (see Section 2.5) and (ii) either syllable or mora is identified as the MPU in a language (see Section 2.4). Then our criteria are meant to assist with steps 2 and 3 of the analytical procedure outlined above. We present two sets of criteria: non-zero-ness criteria (Section 4.2) which help

to distinguish phonologically specified tonal targets from unspecified (default) pitch, and contour criteria (Section 4.3) used to decide when a sequence of tonal targets should be grouped into a contour toneme. Most of these (in both sets) are process-related. However, languages can be found where no tonal processes are attested. For such cases, we have formulated a non-process-related criterion, which is our only general criterion (Section 4.1).

4.1 General criterion

- (8) **Persistence criterion.** If a tone of an m-item that is realized as a single MPU is contrastive and it persists in all contexts (i.e., it is not affected by tonal processes which could be regarded as criteria of tonemic status), it is considered to be a toneme.

This criterion is based on the definition of the toneme and on a general understanding of the nature of linguistic signs. It amounts to saying that if one can isolate a single-MPU m-item (monosyllabic or monomoraic morpheme) with a non-zero tonal specification, that specification is a single toneme.

The single-MPU condition in the formulation of the persistence criterion stems from the need to distinguish between phonologized tonal contours and sequences of level tonemes. The logic behind it is as follows: if we have a lexically specified set of segmental and tonal gestures (m-item), and if this set is minimal in the sense that it can only be realized prosodically as a single co-selection set (one MPU), then all the pitch gestures in this set are assumed to represent a contour toneme. This is clearly just a heuristic to estimate contour entrenchment. But we expect prosodically inseparable gestures in the same modality (pitch) to be highly likely candidates for automatization in both production and comprehension. This criterion is not applicable to m-items that are realized as multiple prosodic units.

For words or morphemes matching the single-MPU condition the criterion excludes the interpretation that one of several tonal targets may be an unspecified, phonologically zero tone. This limitation reflects the fact that realization of a tonal target is an articulatory bottleneck (see Section 2.4), so non-informative (zero) targets will not be attained in a time-constrained context.

This criterion may be sufficient for tonal systems where tones are stable and no (or very few) tonal processes are attested. A well-known example of a language with contour tonemes whose phonological identity can be established mainly by means of the Persistence criterion is Vietnamese. For a non-Asian language with level tones where the Persistence criterion plays a crucial role, one can mention Eastern Dan (Vydryn 2026b [this volume]).

To avoid populating tonal inventories with only marginally attested tonal contours, contour tonemes identified on the basis of the Persistence criterion must also comply with the contour entrenchment condition.

- (9) If a tonal contour is lexically specified for a prosodically minimal morpheme (one MPU m-item), this tonal contour is a toneme *if and only if*:
- a. it is frequent in texts, and / or
 - b. it is productive (i.e. it is frequent in the dictionary or in inflectional paradigms).

This statement provides a heuristic to estimate the degree of automatization (entrenchment) of a contour for the speakers of a language. It combines two conditions that have to be met in order to assume that a lexical contour is well entrenched. First is the prosodic minimality condition (in terms of one MPU), similar to that used by the persistence criterion (8). As evidence for informativity and stability of a tonal contour we use the fact that the contour is retained in prosodically minimal contexts where time pressure is high. But this has to be supported with some data showing that this contour is not idiosyncratic to a particular word, but is attested more widely as a unit in the tonal system. This can be ascertained in two ways listed as subconditions in the formulation of the criterion.

First, where the prosodically minimal items carrying a contour are textually frequent (9a), this is an argument for its tonemic status. This condition is based on the strong correlation of frequency with entrenchment (Divjak 2019: 119). For many languages, there will be no corpora of sufficient size to estimate the frequency directly. But one can rather reliably assume high frequency for certain words on the basis of either their grammatical category (function words are very frequent in all languages) or their semantics (e.g. the word for ‘head’ is likely to be rather frequent). We do not think that it is possible, or even desirable, to set a fixed frequency threshold to apply this condition. It is left to the discretion of the author using the Persistence criterion.

Second, if a contour displays productivity as a distinctive unit in the vocabulary of the language and/or in grammar (9b), we similarly regard it as an argument for the contour being entrenched, and hence a toneme. The productivity may be estimated by the number of lexical or grammatical items in the dictionary that have the given contour as their tonal specification. For this subcondition, lexical items of any prosodic size should be taken into account, not only minimal morphemes consisting of one MPU. As above, we do not suggest a fixed productivity threshold for this condition to apply.

The condition applies whenever either one or both of its subconditions (9a) and (9b) occur.

4.2 Non-zeroness criteria

4.2.1 Status of zero tones in tonology

Since normally all syllables are pronounced with some pitch, the first analytical task is to make sure that certain pitches constitute phonologically specified tonal targets as opposed to default, phonologically zero tones. In earlier works, the problem of zero tones in tonal languages was usually treated in terms of markedness. This is the case for Maddieson, who specified the following criteria of tonal markedness (Maddieson 1978: 341):

- frequency of tones (a less frequent tone is marked);
- dominance in assimilatory processes (a tone which is not changed by assimilation is marked);
- existence of neutralized tones (e.g., in a language where stress is a factor, tones in stressed syllables are marked in relation to their replacements in unstressed positions).

The notion of markedness is rather vague and subject to various interpretations (Haspelmath 2006), for which reason we refrain from using it in our argumentation. If we consider Maddieson's criteria in terms of their fitness to serve as criteria of tonemic status, the following observations can be made:

- 1) Frequency of tones. This criterion cannot be applied before we establish the inventory of tonemes in a language (i.e., before we decide what is or is not a toneme). We are therefore trapped in a vicious circle. But even if we assume that Maddieson meant surface tones, it remains unclear how frequent a tone needs to be in order not to be granted tonemic status.
- 2) Dominance in assimilatory processes. As shown in Hyman (2007a), tonal processes are not limited to assimilation. Nonetheless, this criterion can be seen as a prototype of our Activity criterion (see below, Section 4.2.3).
- 3) Tonal neutralization in this criterion fails to distinguish between two different notions: zero tone (which is not a toneme by definition) and unmarked tone (zero is unmarked, but tonemes can be more or less marked as well). Therefore, it cannot serve as a non-zeroness criterion.

As an alternative to markedness criteria, other authors have searched for criteria to “zero out” one of the tones in the system (in our terminology, to distinguish between a toneme and a tone which does not represent a toneme). This topic has been of special interest for specialists in Bantu languages, where two-level tonal systems are predominant. Some authors embrace an extreme position, such as Stevick (1969: 330): “Whenever a system consists of two contrasting entities, the analyst may suggest an alternative interpretation whereby one of the entities is ‘zeroed out’ and the contrast is regarded as presence vs. absence of some one positive entity.”

In the most authoritative publication in this field, Hyman (2000), a more balanced approach is taken. Hyman recognizes that if a language has two tonal levels its tonal system may be privative (/L/ vs. Ø or /H/ vs. Ø) or binary (i.e., /L/ vs. /H/), and formulates a number of criteria which make it possible to determine whether a tone is zero or not.

Most of our criteria for the tonemic status of a tone are inspired by Hyman’s non-zeroness criteria, reformulated in a way that conforms to our assumptions. The main point of the previous publications was to distinguish what is a “(marked) tone” from what is “not a tone” (or an “unmarked tone” or “zero tone”) in privative tonal systems. We separate this into two distinct problems. One is to differentiate between tonal targets that serve some lexicogrammatical function (tonemes proper) and phonologically neutral tonal fillers (default tone). Another is to represent tonemic markedness as a scalable phenomenon: the more contexts where criteria are met by a toneme, the more marked it is.

The essential idea on which all our non-zeroness criteria are based is that default (zero) pitch is not informative. If some tonal target is realized in a situation of articulatory or time constraints, it is informative and hence not a zero. Similarly, if a tonal target is preserved when displaced relative to sonorous segments of its m-item, it is likewise informative and non-zero.

4.2.2 The shared MPU criterion

- (10) If two level tones (pitch targets) can be realized in one MPU as a result of a tonal process, each tone is a toneme or a part of a toneme, and neither is Ø.

This criterion is based on the assumption that if a tonal target is preserved in a constrained prosodic environment, that target is informative and thus phonologically specified (non-zero). In Hyman’s wording (2000), “it should not be possible to have HL or LH contour tones in a language in which the opposition is /H/ vs.

/Ø/". Hyman also mentions that the failure to have contours with a mid tone as a component was used earlier by Pulleyblank (1986) as a strong argument for the unmarkedness (i.e., zeroness) of Mid in Yoruba.²

A precondition to the shared MPU criterion is that the contour in question is not necessarily a single toneme. Logically, it can represent two level tonemes, a combination of a level toneme with a part of a contour toneme that spans other MPUs, or even a combination of the end and the beginning of two different contours. However, we would not expect the latter to be a common occurrence.

The shared MPU criterion can be extended seamlessly to the case where three level tones are found on one MPU.

4.2.3 The activity criterion

- (11) If a tone is able to surface outside its host m-item or triggers a tonal change outside its host m-item, it is a toneme or a part of a toneme.

The activity criterion can be regarded as a modified version of Hyman's (2000) "tonal rule feature": "If the opposition is /H, Ø/, tone rules should refer only to H's", and of Maddieson's criterion of "dominance in assimilatory processes" as mentioned above.

In our terms, the activity criterion boils down to the statement that only tonemes, but not zero tone, can cause tonal alternations in neighboring m-items. The first two classes of tonal processes are considered here: either when a tonal target is realized outside the m-item to which it belongs, or when the presence of a tonal target causes a change to a tonal target in another m-item, e.g. polarization.

There are a few typical situations where the activity criterion can be applied. One is tonal spreading to the right as, for example, in Bambara (12). The lexical tone of *jùru* is low, but in the compound it is integrated into the tonal span of the H tone of *dá*. The presence of H on *dá* is the reason to suppress the L of *jùru*. This is evidence for the tonemic status of H according to the activity criterion.

- (12) *dá* 'mouth', *jùru* 'rope' → *dá juru* [H *dá júru*] 'rein'

In contrast, a similar tonal alternation may not be informative with respect to tonemic status when there is no evidence for the lexical tone of the m-item, as is commonly the case with bound morphemes. For example, in Bambara the intransitive perfective marker *-ra/-la* always takes the tone of the stem (13). At the

²Later on, Pulleyblank (2004) came to the conclusion that M, although ranking lower than H and L in the hierarchy of markedness, is not phonologically unspecified, i.e., it is not zero tone.

starting point of the analysis of the tonal system no tonemes are yet identified. In this case, it is conceivable that the system is privative, and either H or L could be a zero tone. Then the activity criterion is inapplicable, as H and L behave symmetrically with respect to the tonalization of *-ra/-la*. The affix is best analyzed as toneless.

- (13) a. déli ‘to pray’ + *-ra* → délí-lá
 b. dèli ‘to get accustomed’ + *-ra* → dèli-là

Tonal morphemes constitute another common case for the activity criterion. In our terms, whenever we can isolate an m-item which has a zero segmental specification and a non-zero tonal specification (a tonal morpheme) the tonal specification must contain at least one toneme.

The simplest tonal morpheme consists of just one tonal target (without a segmental part). For example, in Eastern Dan, the grammatical morpheme expressing neutral aspect is an extra-low tone that replaces a lexical tone on the verbal stem (14).

- (14) Eastern Dan
 Gō yǝ kāl dǎŋ\ xL bǝǝ gú.
 [M gō] [xL yǝ] [M kāl] <[xL dǎŋ] > [M bǝǝ] [xL .ǝ] [H gú]
 vehicle 3SG.EXI RETR stick\NEUT mud in
 ‘A car got stuck in the mud.’

In our terms, we would call the neutral aspect marker in Eastern Dan an m-item with zero segmental specification, or simply a tonal morpheme. Since xL surfaces on the MPUs of other m-items it falls under our first type of tonal process. Hence the xL tonal target which is the only exponent of this morpheme must be a toneme according to the Activity criterion (11).

A tonal morpheme by definition does not have segments of its own, and the tonal targets have to surface on prosodic units formed by other m-items. Thus they will necessarily project outside their host m-item (thereby participating in a process), the only reason for that being that it is informative (non-zero). Some grammatical morphemes combine both segmental and tonal exponents, where the tonal part of the exponent is realized on another morpheme, typically a stem. The latter case similarly fits the activity criterion.

A tonal morpheme can have a complex structure, in the sense that it consists of two or more tonal targets. Complex tonal morphemes are far from rare. They are attested in Soninke, Mwan, Hausa and Jamsay, to name just a few. For instance,

in Soninke a LH morpheme marks an adjective as having predicative function (15).

- (15) Soninke
 Ké wúlle qúllè-n\LH nì
 [H ké] [H wúllé] <[L qù][H llé-n]> [L nì]
 this dog white-DEF\STCONSTR EQU
 ‘This dog is white’

For a complex tonal morpheme, we exclude the interpretation that any of its tonal targets could be a zero tone.

There is no logical requirement that each tonal target in a complex tonal morpheme must have a tonemic status individually. But this is not logically excluded either, and should be verified with reference to contour criteria³.

Some kinds of alternations traditionally described as involving floating tones also fit under the definition of the activity criterion. For example, in Babanki the hodiernal past tense marker $tə^L$ has a second low tone that is realized on the first syllable of the following high-toned verb (16).

- (16) Babanki
 Mà tə^L sàŋlǎ zòjn.
 [L mà] [L tə] [L sàŋ][H lǎ] [L zòjn]
 1SG P2 rejoice yesterday
 ‘I rejoiced yesterday.’

4.2.4 The intonational criterion

- (17) If a tonal target is maintained in an interaction with an intonational pitch/contour, either at the cost of modifying or failing to realize the intonational tone, or by means of a shift of a lexical tonal target to a different segmental position, this tonal target is a toneme or a part of a toneme.

The interaction of tone and intonation is a complex topic, with significant variation across tonal languages. As both lexical tone and intonation are realized in the same articulatory modality of pitch control, they are sometimes in competition with each other. This often happens at the phrase end when lexical tones

³In a language where both /H/ and /L/ have tonemic status, like Soninke, we would not postulate a /LH/ toneme solely on the basis of a complex tonal morpheme with such a contour. We prefer a composite interpretation m(Ø; /L/ /H/) in the absence of independent data supporting the contour as an item in the tonal inventory.

interact with intonational boundary tones. One of the possibilities attested in tonal languages is to maintain a lexical tonal contrast at the expense of the realization of the intonational tone. If the local density of tones permits it, a lexical tone can be maintained by shifting it to a different segmental position. In both cases, the maintained tonal target is clearly phonologically specified and is not a zero tone. Our criterion targets all situations of this kind.

4.3 Contour criteria

The next step after the identification of the tonal targets is to determine whether some of them form phonological tonal contours, and hence need to be grouped into contour tonemes. It is important to highlight that our understanding of a toneme is based on its distinctive function, and not on the surface realization. Thus for any sequence of tonal targets to be considered a contour toneme, all of the targets in the contour must belong to the same morphological entity (m-item). This means that the attestation of a tonal contour on some, even minimal, prosodic unit (MPU) is *by itself* insufficient to claim the presence of a contour toneme. At the same time, if a sequence of tonal targets belongs to the same lexical item (m-item), this is also not sufficient reason to treat it as a single toneme. An exponent of a morpheme may comprise multiple tonemes, just as it may comprise multiple segments. Other sources of evidence are thus required to support the interpretation that a contour actually functions as a single phonological unit.

We based the diagnostic criteria for contour tonemes suggested below on the explicit recognition of the morphological nature of the toneme. Each criterion combines recognition of the distinctive function of a tonal contour on a single m-item with some heuristic allowing one to infer its phonological unity. Most often, we will have no direct way to assess the degree of entrenchment of a contour for the speakers of the language. Therefore we resort to using heuristics that are based on data likely available in the descriptive literature for most languages.

4.3.1 Non-compositionality criterion

- (18) If a tonal contour of an m-item is realized on one MPU, and at least one of the tonal targets of the contour is not available in this language as a toneme, this contour is a toneme.

This criterion proceeds from the general properties of the linguistic sign. It follows a simple logic: if a component of a signifier does not exist as a separate sign, the signifier itself is a single sign. This logic was applied to the identification of

phonological units by Trubetzkoy (1969). In other terms, this is a limiting case of entrenchment, whereby one or all of the components comprising the contour completely lose the ability to be used autonomously, and are thus transformed into a complex unit – a contour toneme. This criterion is crucial for those languages where tonal processes are lacking or very marginal.

The non-compositionality criterion was widely used by Pike (1948); it was also discussed at length by Yip (2002: 50-51). Our only modification is to limit its application to the context of a single m-item spanning just one MPU. The limitation to one MPU reflects the cognitively plausible assumption that pitch gestures coordinated within an elementary prosodic unit will be highly entrenched, and hence produced and perceived as a single unit. The single MPU condition precludes us from always treating a lexically specified tonal contour as one toneme (see Section 5.3 for a comparison with the morphotoneme).

By contrast, the reverse is not true: if the levels which constitute a contour on an m-item that spans one MPU are available in a language as tonemes, this does not necessarily mean that the contour is not a toneme. The tonemic or non-tonemic nature of the contour must be checked according to the other criteria.

4.3.2 The holistic criterion

- (19) If a tonal specification of an m-item consists of a sequence of tonal targets, and *none* of the targets can be shown to constitute a toneme in the language, this sequence is a toneme.

This criterion is based on the same idea of non-compositionality as that discussed in connection with the non-compositionality criterion (Section 4.3.1). But in contrast to the formulation in (18), the holistic criterion can be applied to lexical items (m-items) that are not prosodically minimal. The holistic criterion is essential to identify tonemes in languages with low tonal density and contour tonemes, often referred to as pitch accent or melodic stress languages. In our sample these are Iquito and Japanese.

This criterion is made consistent with the assumptions of our framework by the requirement that none of the tonal targets in a sequence be an independent toneme. This ensures that the interpretation of one of the tonal targets as a default tone remains theoretically possible. For example in a privative system of /H/ vs. Ø, the surface tonal contour of a long lexical item may appear as HL, LH or LHL, but only one /H/ toneme is present in all cases.

5 Discussion

5.1 Rejected criteria

Our primary goal was to systematize the criteria and principles involved in identifying the units of tonal inventories, which we call tonemes, in typologically diverse tonal languages. We aimed to formulate our criteria for toneme identification in such a way that they rely only on the explicitly made assumptions discussed in Section 2 and Section 3, and do not involve other theoretical constructs. This resulted in some revisions to the corpus of diagnostic criteria found in previous literature.

First, we adapted some of the existing criteria, recasting them in our terms. Both the shared MPU criterion (10) and the activity criterion (11) were inspired by the ideas formulated by Hyman (2000). Similarly, we use a restricted version of the non-compositionality criterion (18) for contour tonemes used by Pike and other tonologists.

Second, we formulated several new criteria based on our set of assumptions. These new criteria mostly stem from ideas that were already “in the air” but had not been explicitly formulated as diagnostic criteria for tonal inventories. This in fact applies to the persistence criterion (8). Arguably, the more novel criteria that have not been explicitly formulated before are the intonational criterion for non-zereness (17), together with the holistic criterion (19) for contour tonemes.

Finally, some criteria are found in the literature that we have rejected or at least suspended on theoretical grounds. The main reason for doing so was their inconsistency with the assumptions we have chosen to make. It seems appropriate to give these criteria some discussion.

5.1.1 The floating criterion

In discussing privative tone in Bantu, Hyman (2000) suggested a criterion based on the notion of floating tone. In Hyman’s wording, “only specified tones can float”. An alternative formulation of this criterion is: A zero tone cannot float. In our terms, this is equivalent to stating that if a tonal target can float, it is a toneme or a part of a toneme. In our view, this criterion spans several structurally different but nevertheless logically valid cases.

The main problem with this criterion for us lies with the notion of floating tone. Floating tone is commonly postulated in the autosegmental tonology as an abstract entity which helps to explain irregularities of the surface tonal distribution. These irregularities can be thought of as coming from the historical

presence of some phonological tone, or from a synchronic phonological process involving underlying tones. Over time, floating tone grew into a kind of umbrella term encompassing quite disparate phenomena.

Autosegmental theory reclaimed the heuristic notion of floating tone as its own theoretical construct. Goldsmith (1976: 45-50) understood a floating tone to be a tone that is not underlyingly associated to a TBU. The same idea is echoed in the recent reformulation by Rolle and Lionnet (2020), which extends the notion of floating tones to cover tones not associated to a TBU at some moment in the derivational process: “floating tones are defined as tones in a representation which are not associated to a tone-bearing unit (TBU). This representation may be either part of the underlying form of a morpheme, or arise at some derivational stage.” But although alternative definitions of floating tone all stem from the same theoretical idea of autosegments, the exact set of phenomena labeled as floating tones differ significantly in the analyses offered by different authors. Additionally, Rolle and Lionnet’s definition presupposes that the scope of the concept depends on the derivational processes posited, not on empirically observed characteristics. In our system, where we try to minimize assumptions about abstract representations, this is an unfortunate feature. In the limit, the idea of autosegments presumes that at some initial point in the derivation all tones are unassociated, and hence all tones are floating by the letter of this definition.⁴

Yet the surface alternations targeted by the concept of floating tone still provide evidence relevant to our task of toneme identification. The problem is that without the assumptions of the autosegmental framework this set of alternations loses theoretical coherence and appears to be an assemblage of quite disparate phenomena. When translated into our terms, all of this evidence is subsumed under different existing criteria, meaning that an additional criterion is unnecessary.

Below, we give an example of the kind of alternation that is commonly interpreted in terms of floating tone, and indicate how it is covered by our activity criterion.

In Mundurucu (< Eastern Tupian < Tupian), some toneless morphemes have a suffixed floating H as a part of their specification. It surfaces as H on the previous (20a) or following (20b) syllable belonging to the same phonological word, provided that this syllable is word-final and is not preceded by another H-toned syllable (Gerasimov 2024).

- (20) a. /dadʒeʔ^H/ → d̪àd̪ʒé ‘peccary’

⁴We would like to thank Dmitri Gerasimov for sharing this observation with us.

b. /daɖɛʔ^H/ ‘peccary’ + /yã/ PL → dàɖɛ̀=yã ‘peccaries’

When H surfaces on a syllable that belongs to the neighboring morpheme, in this case the plural marker *yã*, this is direct evidence for its non-zero status. The activity criterion (11) applies here. But this fact is not sufficient to determine that H is a standalone toneme, as in *m*(da.ɖɛʔ; Ø.Ø./H/). Other interpretations remain logically possible: in particular, H could be a part of a rising contour toneme assigned to the word *m*(daɖɛʔ; /LH/). Thus the tonemic status of the first L on the stem ‘peccary’ is left open in this example: it could be either a zero or a toneme, and the activity criterion has nothing to say about it. The only interpretation logically excluded by the floating position of H is that L is a toneme and H is zero, **m*(da.ɖɛʔ; /L/Ø).

5.1.2 Distribution criterion

Hyman (2000) provides examples from Bantu languages where the surface appearance of H tone is positionally restricted. For example, in Haya H cannot appear immediately before or after a pause, while there is no restriction on the position of L, for which reason L is interpreted as Ø. If we follow Hyman’s logic, we could formulate this criterion as follows: “There can be constraints on the occurrence of a toneme, while there are no constraints on the occurrence of Ø.”

The problem with this criterion for us is twofold. First, there are too many degrees of freedom in how a constraint could be defined. It would be a large backdoor compromising the cross-linguistic comparability of tonal inventories which we strive for. But however defined, constraints always rely on the generalization of unattested examples to the level of abstract prohibitions. This runs counter to our general principle that tonemic status must be claimed only on the basis of some positive evidence in its favor. Using negative evidence in the form of a suggested constraint seems to us to be logically inconsistent with this principle.

5.1.3 Locality criterion

Another idea used by Hyman (2000) as a test for the non-zero phonological status of a surface tone stems from an assumption as to what constitutes a possible tonal rule. In Hyman’s words: “if the opposition is /H, L/, H tones should not be expected to interact with each other across L’s. If the opposition is /H, Ø/, on the other hand, the H’s on each side of a toneless TBU should be able to ‘see each other’—even across long distances.” This boils down to the strict locality of toneme interactions: only rules that involve pairs (digrams) of neighboring

tonemes are allowed. Given the accumulating data that phonological processes are most of the time strictly local in the relevant sense (Heinz 2018), this seems a plausible theoretical assumption about tonal processes.

Nonetheless, in a discussion of this idea with the present authors, Hyman himself (p.c.), mentioned as a counterexample Aghem (Bantu), a two-level language where both H and L have tonemic status. As shown in Hyman (1987), this language has the following rule: in a sequence Low – High – floating Low, the floating L is deleted. In other words, the deletion of the floating L is conditioned by the presence of a preceding L which is not in direct contact with it.

Another piece of counterevidence comes from the Kugama language (Cameroon-Ubangian) where in a sequence High – floating Low – High – floating Low, the second floating Low is deleted (Litvinova 2026 [this volume]), e.g.:

- (21) $\acute{\text{b}}\text{ĩt}\acute{\text{ɔ}}$ $^{\text{h}}\text{n}\acute{\text{ɔ}}\acute{\text{ɔ}}$ $\text{lé}\bar{\text{y}}$
 $|\acute{\text{b}}\text{ĩt}\acute{\text{ɔ}}^{\text{L}}$ $\text{n}\acute{\text{ɔ}}\bar{\text{ɔ}}^{\text{L}}$ $\text{lé}\bar{\text{y}}$
 SG.1PL.POSS eye=GEN fire
 ‘our gun’ (lit. our eye of fire)

As a result, we refrain for now from including the locality assumption in our framework. The possible near-universal locality of tonal processes remains a highly intriguing topic for further research.

5.1.4 Copy/spread criterion (Yip)

In her discussion of the status of tonal contours, Yip mentions (2002: 51, 54–55) that “in some languages contours seem to copy or spread as units” (in her book Yip points to a few languages where this process occurs: Lalana Chinantec, Changzhi, Danyang Chinese). She sees this process effectively as evidence for the phonological status of such contours as atomic units. While it is tempting to turn this idea into a criterion for contour tonemes, we decided to reject it, for several reasons.

The first process mentioned by Yip, tonal spreading, is described in our terms as the extension of the tonal span of a toneme. Such an expansion necessarily involves additional segmental material, so that a toneme has to span over a different m-item. This satisfies the conditions of the activity criterion (11). But as we have seen above during the discussion of the floating criterion (20), in cases of tonal spreading not all tonal targets of the contour may cross morpheme boundaries. Then the possibility that the contour is composed of several tonemes, or even of a default tone and a toneme, is not logically excluded solely by the fact

of the spread. This leaves tonal spread as a criterion for tonemic status partly redundant, and partly still insufficient to argue for the phonological status of a contour.

The second process, labeled by Yip as copying, may be described in our terms as the creation of a new toneme equivalent to the original one. But in this case we see no reason why non-monotonemic contours cannot be copied. In fact, examples of multi-tonemic contour copying can easily be found. For instance, in Mano (Mande), a language with three level tonemes (/H/, /M/, /L/), some adjectives derive their plural or plural intensive forms through full reduplication, and the bitonemic tonal contour is copied onto the reduplicand, e.g.: *gbókò* ‘numerous; thick’ → pl. *gbókò-gbókò* (Khachaturyan 2014: 59).

5.2 Caveats and corner cases for our system of criteria

5.2.1 Articulatory blending of pitch gestures

Pitch gestures that form the backbone for tonal contrasts are subject to the same articulatory forces that lead to reduction, assimilation and coarticulation. Contexts where tonal targets are not clearly separable in the phonetic realization may result from these processes. The application of our criteria in such contexts may be complicated, as they rely on the identification of the tonal targets to infer tonemic status.

The most cross-linguistically common case of tonal coarticulation relevant to our system is non-automatic downstep.⁵ In general, downstep is a register operation that shifts down the ceiling of the speaker’s pitch range until the end of the phrase. The register typically goes down automatically after a realized low tone. But in some contexts the downstep of the following high tone is the only overt indication of the presence of a low tone, when the low pitch target is not itself realized. These situations are usually described in the literature as instances of a floating low tone. For our analysis, the presence of a non-automatic downstep is clearly informative with respect to the tonemic status of the low pitch target, as information about its presence is preserved in the form of the downstep. Downstep can be interpreted as an allotone of the low toneme in these instances, realized at the moment when the following high pitch target starts. Acoustically, only one pitch target is realized, a downstepped high. But functionally, the presence of both low and high targets is communicated simultaneously in this realization.

⁵We do not consider here cases where downstep serves as a boundary signal, as in Lhasa Tibetan (Perekhvalskaya 2026 [this volume]) or Wanga (Marlo & Green n.d.), or as a self-standing toneme, as in Pukur (Rochant 2023) or Paicî (Lionnet 2022).

A similar concern may be raised in cases where coarticulation of tonal targets in a constrained environment causes undershoot that results in a pitch target that is distinct from both blending targets. This is the case in Babanki, where the phonological sequence of H and L coinciding on a single syllable is realized as M (Akumbu & Vydrin 2024). One may somewhat artificially describe this context in our terms supposing that M serves as an allotone of both H and L tonemes in Babanki. The application of the shared MPU criterion is debatable in this case.

Our system of criteria does not currently cover the above cases, but we still see these data as relevant for inferring the tonemic status. The shared MPU criterion stands closest to the treatment of the coarticulated tonal targets discussed in this section. The general idea behind the shared MPU criterion that gives it validity as a diagnostic for tonemic status is preservation of the information in the articulatory constrained environment. This idea is also applicable to the cases just mentioned.

5.2.2 Subtractive tonal morphemes

Another question concerns subtractive grammatical tonal morphemes. A subtractive morpheme must manifest itself as the grammatically meaningful deletion of tonemes. In our notation, it can be presented as $m(-T; \emptyset)$. Still, one needs good reasons to postulate morphemic status for the deletion operation, i.e., to classify it as a morpheme-operation.⁶ If we were to admit that removal of a toneme is the exponent of such a morpheme, that could be indirect evidence for the tonemic status of the deleted tone. But in general, to minimize the assumptions required we would rather not invoke subtractive tonal morphemes as primary evidence for tonemic status.⁷

5.3 Toneme vs. morphotoneme and underlying tone

Toneme is an old term still current in tonology nowadays, albeit not that popular. Given the long and involved history of its usage, too little effort has been spent on defining it. In using the term *toneme*, we refer to its structuralist core – a distinctive function. It also lets us honor the intuitions accumulated in the century-long history of tonology. But simultaneously the use of this term inevitably creates

⁶See Sande (2022) for criteria to distinguish between item- and process-based interpretations of tonal grammar.

⁷See discussion in Vydrin (n.d.) about the interpretation of the tonal system of Somali. We find more convincing the explanation which does not require the introduction of subtractive tonal morphemes in this language.

confusion as to how it relates to more mainstream phonological concepts about the representation of tone.

One possible confusion is with the notion of the morphotoneme. In principle, this term mirrors the analogous term for segmental units – morphophoneme, or m-phoneme, an input segment in the specification of a morpheme (Kiparsky 2018). Since our definition of toneme relies on the morphological understanding of tone (Section 2.2), it stands rather close. However in the tonological literature the term morphotoneme is often used in a different sense, as a synonym for tonal morpheme (e.g. Creissels 2015). Following this latter understanding, any tonal contour associated with a morpheme would be a single morphotoneme. But the crucial difference is that we do not assume that a morpheme can have only one toneme in its exponent. We call such a contour a tonal melody that may consist of several tonemes.

One may also wonder why it is worth using the term *toneme* when *underlying tone* is commonly accepted for discussion of tonal items in the phonological inventory. We avoid talking of underlying tones because of the term's representational connotations. *Underlying tone* is highly embedded in the autosegmental and, more widely, the generative tradition in phonology. As such, it brings with it multiple assumptions originating in those phonological frameworks. For instance, in comparison to our notion of the toneme, an underlying tone is typically restricted to a single tonal target (contours are not considered to be possible phonological units). In pursuit of our typological goals, we aim to limit the number of assumptions made on the representational side of things. To communicate our intention, we selected an alternative term *toneme* which is defined, first and foremost, through structural relations between surface tonal realizations.

6 Conclusion

We present a theoretical framework developed for the typological study of tonal systems in the languages of the world. The path we have taken is to minimize the assumptions on the representation of tone, keeping our concepts motivated by either surface phenomena or function. Very briefly, these include the toneme as a minimal distinctive unit implemented with pitch gestures, the idea that tonemes are coordinated with segments within some prosodic units, and the understanding of tonal processes as alternations in toneme-segment coordination.

In this article, we have focused specifically on the problem of identifying the inventory of tonemes in a language, and suggest a set of diagnostic criteria to help proceed from surface distributional phenomena to a set of items in the

tonal inventory. On the practical side, our set of criteria is limited by the fact that for the majority of languages in a typological survey neither corpora nor experimental studies of tone will be available. Our criteria are thus designed keeping in mind that they should be operational given the data typically available in descriptive grammars. While it is probable that our system of criteria can be extended in the future, in practice it is sufficient to identify tonemes in a diverse sample of tonal languages represented in this volume.

Abbreviations and glosses

Symbols and diacritics for tonal levels

xH	ǎ	extra high
H	á	high
M	ā	mid
L	à	low
xL	ǎ	extra low

Glosses and other conventions

1, 2, 3 etc.	numbers of noun classes in Bantu
AM	associative marker
ART	article
GEN	gentive
m	m-item
MPU	minimal prosodic unit
LOC	locative
PST.HOD	hodiernal past
PL	plural
POSS	possessive
SG	singular
TBU	tone-bearing unit

References

- Akumbu, Pius & Valentin Vydrin. 2024. *Babanki. analytical report on the tonal system* (ThoT Project Reports 7). Villejuif. DOI: [10.5281/zenodo.14725438](https://doi.org/10.5281/zenodo.14725438).
- Arvaniti, Amalia, Argyro Katsika & Na Hu. 2024. Variability, overlap, and cue trading in intonation. *Language* 100(2). 265–307.

- Arvaniti, Amalia, D Robert Ladd & Ineke Mennen. 1998. Stability of tonal alignment: The case of Greek prenuclear accents. *Journal of phonetics* 26(1). 3–25.
- Arvaniti, Amalia, D Robert Ladd & Ineke Mennen. 2006. Tonal association and tonal alignment: Evidence from Greek polar questions and contrastive statements. *Language and speech* 49(4). 421–450.
- Beach, Douglas M. 1923. The science of tonetics and its application to bantu languages. *Bantu Studies* 2(1). 75–106.
- Bertet, Denis. 2021. Tikuna, a ten-toneme language in Amazonia. *Amerindia* 43. 55–101. DOI: [10.56551/VHJB2810](https://doi.org/10.56551/VHJB2810).
- Blumenthal-Dramé, Alice. 2012. *Entrenchment in usage-based theories: What corpus data do and do not reveal about the mind*. Walter De Gruyter.
- Browman, Catherine P & Louis Goldstein. 1992. Articulatory phonology: An overview. *Phonetica* 49(3-4). 155–180.
- Burroni, Francesco. 2023. *Dynamics of f0 planning and production: Contextual and rate effects on Thai tone gestures*. Cornell University. (Doctoral dissertation).
- Burroni, Francesco & Sam Tilsen. 2025. Thai speakers time lexical tones to supralaryngeal articulatory events. *Journal of Phonetics* 108. 101389.
- Clements, George N., Alexis Michaud & Cédric Patin. 2011. Do we need tone features? In *Tones and features: Phonetic and phonological perspectives*. John A. Goldsmith, Elizabeth Hume & Leo Wetzels (eds.). Berlin, Boston: De Gruyter Mouton. 3–24. DOI: [10.1515/9783110246223.3](https://doi.org/10.1515/9783110246223.3).
- Coupe, Alexander R. 2014. Strategies for analyzing tone languages. *Language Documentation & Conservation* (8). 462–489.
- Creissels, Denis. 2015. The inflectional role of tone in soninke (west mande). In *45th colloquium on african languages and linguistics*. Leiden.
- d’Imperio, Mariapaola, Robert Espesser, Hélène Loevenbruck, Caroline Menezes, Noël Nguyen & Pauline Welby. 2007. Are tones aligned with articulatory events? evidence from Italian and French. *Papers in Laboratory Phonology* 9. 577–608.
- Divjak, Dagmar. 2019. *Frequency in language: Memory, attention and learning*. Cambridge University Press.
- Gao, Man. 2008. *Tonal alignment in Mandarin Chinese: An articulatory phonology account*. Yale University. (Doctoral dissertation).
- Gerasimov, Dmitry. 2024. *Munduruku. analytical report on the tonal system* (ThoT Project Reports 10). Villejuif.
- Gerasimov, Dmitry. 2026. Dom tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 465–502. Berlin: Language Science Press. DOI: [??](https://doi.org/10.56551/VHJB2810).

- Goldsmith, John A. 1976. *Autosegmental phonology*. Cambridge, MA: MIT. (Doctoral dissertation).
- Gordon, Matthew Kelly. 2001. A typology of contour tone restrictions. *Studies in Language* 25. 405–444.
- Gordon, Matthew Kelly. 2016. *Phonological typology* (Oxford surveys in phonology and phonetics 1). Oxford: Oxford University Press.
- Gouskova, Maria & Juliet Stanton. 2021. Learning complex segments. *Language* 97(1). 151–193.
- Haspelmath, Martin. 2006. Against markedness (and what to replace it with). *Journal of Linguistics* 42(1). 25–70. DOI: [10.1017/S0022226705003683](https://doi.org/10.1017/S0022226705003683).
- Haspelmath, Martin. 2023. On what a construction is. *Constructions* 15(1).
- Haspelmath, Martin. 2024. Four kinds of lexical items: Words, lexemes, inventorial items, and mental items. *Lexique. Revue en Sciences du Langage* (34).
- Heinz, Jeffrey. 2018. The computational nature of phonological generalizations. *Phonological typology, phonetics and phonology* 23. 126–195.
- Hu, Fang. 2016. Tones are not abstract autosegmentals. In *Speech prosody*, vol. 2016, 302–306.
- Hyman, Larry M. 1987. Downstep deletion in aghem. In David Odden (ed.), *Current approaches to african linguistics*, vol. 4, 209–222. Dordrecht: Mouton de Gruyter.
- Hyman, Larry M. 2000. Privative tone in bantu. In *Symposium on tone, ilcaa*. Tokyo.
- Hyman, Larry M. 2007a. Universals of tone rules: 30 years later. In Tomas Riad & Carlos Gussenhoven (eds.), *Tones and tunes: Typological studies in word and sentence prosody*, vol. 1, 1–34. Berlin, New York: Mouton de Gruyter. DOI: [10.1515/9783110207569.1](https://doi.org/10.1515/9783110207569.1).
- Hyman, Larry M. 2007b. Where's phonology in typology? *Linguistic Typology* 11(1). DOI: [10.1515/LINGTY.2007.021](https://doi.org/10.1515/LINGTY.2007.021).
- Hyman, Larry M. 2011a. Do tones have features? In *Tones and features: Phonetic and phonological perspectives*. John A. Goldsmith, Elizabeth Hume & Leo Wetzels (eds.). Berlin, Boston: De Gruyter Mouton. 50–80. DOI: [10.1515/9783110246223.50](https://doi.org/10.1515/9783110246223.50).
- Hyman, Larry M. 2011b. Tone: Is it different? In John A. Goldsmith, Jason Riggle & Alan Yu (eds.), *The handbook of phonological theory*, 2nd edn., 197–239. Blackwell.
- Hyman, Larry M. & William R. Leben. 2021. Tone systems. In Carlos Gussenhoven & Aoju Chen (eds.), *The oxford handbook of language prosody*, 45–65. Oxford.

- Hyman, Larry M. & Russell G. Schuh. 1974. Universals of tone rules: Evidence from West Africa. *Linguistic Inquiry* 5(1). 81–115.
- Kaldhol, Nina Hagen. 2024. A typology of tonal exponence. *Morphology* 34. 321–367. DOI: [10.1007/s11525-024-09427-w](https://doi.org/10.1007/s11525-024-09427-w).
- Kang, Weiyi & Yi Xu. 2024. Tone-syllable synchrony in Mandarin: New evidence and implications. *Speech Communication* 163. 103–121.
- Karlin, Robin. 2022. Expanding the gestural model of lexical tone: Evidence from two dialects of Serbian. *Laboratory Phonology* 13(1).
- Karlin, Robin & Sam Tilsen. 2014. The articulatory tone-bearing unit: Gestural coordination of lexical tone in Thai. In *Proceedings of meetings on acoustics*, vol. 22, 060006.
- Karlin, Robin Park. 2018. *Towards an articulatory model of tone: A cross-linguistic investigation*. Cornell University. (Doctoral dissertation).
- Khachaturyan, Maria. 2014. *Grammaire de la langue Mano (Mandé-Sud) dans une perspective typologique*. Paris: INALCO. (Doctoral dissertation).
- Kim, Seung-Eun & Sam Tilsen. 2025. The gesture-field-register (gfr) framework for modeling f0 control. *Journal of Phonetics* 110. 101410.
- Kiparsky, Paul. 2018. Formal and empirical issues in phonological typology. In Larry M. Hyman & Frans Plank (eds.), *Phonological typology*, 54–106. Berlin: De Gruyter Mouton.
- Kirby, James & Marc Brunelle. 2017. Southeast asian tone in areal perspective. In Raymond Hickey (ed.), *The cambridge handbook of areal linguistics*, 703–731. Cambridge: Cambridge University Press.
- Ladd, D Robert, Dan Faulkner, Hanneke Faulkner & Astrid Schepman. 1999. Constant “segmental anchoring” of f movements under changes in speech rate. *The Journal of the Acoustical Society of America* 106(3). 1543–1554.
- Ladd, D Robert, Ineke Mennen & Astrid Schepman. 2000. Phonological conditioning of peak alignment in rising pitch accents in Dutch. *The Journal of the Acoustical Society of America* 107(5). 2685–2696.
- Langacker, Ronald W. 1987. *Foundations of cognitive grammar: Volume i: Theoretical prerequisites*, vol. 1. Stanford University Press.
- Lionnet, Florian. 2022. Tone and downstep in Paici (Oceanic, New Caledonia). *Phonological Data and Analysis* 4(1). 1–47. DOI: [10.3765/pda.v4art1.45](https://doi.org/10.3765/pda.v4art1.45).
- Litvinova, Lora. 2024. *Analytical report on the hausa tonal system*. <https://thot.huma-num.fr/languages/>.
- Litvinova, Lora. 2026. Kugama tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 299–327. Berlin: Language Science Press. DOI: ??.

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- Lundmark, Malin Svensson, Johan Frid, Gilbert Ambrazaitis & Susanne Schötz. 2021. Word-initial consonant–vowel coordination in a lexical pitch-accent language. *Phonetica* 78(5-6). 515–569.
- Maddieson, Ian. 1978. Universals of tone. In Joseph H. Greenberg (ed.), *Universals of human language*, vol. 2, 335–365. Stanford: Stanford University Press.
- Maddieson, Ian. 2013. Tone (v2020.4). In Matthew S. Dryer & Martin Haspelmath (eds.), *The world atlas of language structures online*. Zenodo. DOI: [10.5281/zenodo.13950591](https://doi.org/10.5281/zenodo.13950591).
- Maddieson, Ian. 2023. Investigating the ‘what’, ‘where’ and ‘why’ of global phonological typology. *Linguistic Typology* 27(2). 245–266. DOI: [10.1515/lingty-2022-0076](https://doi.org/10.1515/lingty-2022-0076).
- Marlo, Michael R. & Christopher R. Green. N.d. Wanga: An analytical report on the tonal system. FIXME: preprint reference incomplete — Wanga did not appear in this volume; supply the ThoT analytical report / Zenodo DOI.
- Maslinsky, Kirill, Valentin Vydrin & Dmitry Gerasimov. 2024. *ThoT Database*. <https://thot.huma-num.fr/db/>.
- Morin, Cameron. 2025. Are phonemes constructions? a plea for distinguishing function and meaning. *Constructions and Frames*.
- Mücke, Doris, Hosung Nam, Anne Hermes & Louis Goldstein. 2012. Coupling of tone and constriction gestures in pitch accents. *Consonant clusters and structural complexity* 26. 205.
- Myers, Scott. 1999. Tone association and f_0 timing in Chichewa. *Studies in African linguistics* 28(2). 215–240.
- Perekhvalskaya, Elena. 2026. Lhasa Tibetan tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 441–461. Berlin: Language Science Press. DOI: ??.
- Pike, Kenneth Lee. 1948. *Tone languages: A technique for determining the number and type of pitch contrasts in a language, with studies in tonemic substitution and fusion* (University of Michigan Publications. Linguistics 4). Ann Arbor.
- Polyakova, Eva. 2026. Japanese tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 397–439. Berlin: Language Science Press. DOI: ??.
- Pulleyblank, Douglas. 1986. *Tone in lexical phonology*. Dordrecht: Reidel.
- Pulleyblank, Douglas. 2004. A note on tonal markedness in Yoruba. *Phonology* 21(3). 409–425.
- Rochant, Neige. 2023. *A bilectal grammar of Baga Pukur, an Atlantic language of Guinea*. Paris: Sorbonne Nouvelle. (Doctoral dissertation).

- Rolle, Nicholas & Florian Lionnet. 2020. Phantom structure: A representational account of floating tone association. In *Proceedings of the 2019 annual meeting on phonology*. Linguistic Society of America.
- Sagey, Elizabeth. 1988. On the ill-formedness of crossing association lines. *Linguistic Inquiry* 19(1). 109–118.
- Sande, Hannah L. 2022. Is grammatical tone item-based or process-based? *Phonology* 39(3). 399–442. DOI: [10.1017/S0952675723000106](https://doi.org/10.1017/S0952675723000106).
- Shaw, Jason A, Sejin Oh, Karthik Durvasula & Alexei Kochetov. 2021. Articulatory coordination distinguishes complex segments from segment sequences. *Phonology* 38(3). 437–477.
- Stevick, Earl W. 1969. Tone in bantu. *International Journal of American Linguistics* 35. 330–341.
- Tilsen, Sam. 2014. Selection and coordination of articulatory gestures in temporally constrained production. *Journal of Phonetics* 44. 26–46.
- Trubetzkoy, Nikolai Sergeevich. 1969. *Principles of phonology*. Berkeley: University of California Press.
- Vydrin, Valentin. 2026a. Bambara tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 117–160. Berlin: Language Science Press. DOI: [??](#).
- Vydrin, Valentin. 2026b. Eastern Dan tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 161–199. Berlin: Language Science Press. DOI: [??](#).
- Vydrin, Valentin. N.d. To the interpretation of the somali tonal system. FIXME: preprint reference incomplete — Somali did not appear in this volume; supply the analytical report / Zenodo DOI.
- Vydrin, Valentin & Jeffrey Heath. 2026. Bunoge tonal system. In Valentin Vydrin, Dmitri Gerasimov & Kirill Maslinsky (eds.), *Parameters of tonal systems: Volume 1*, 265–297. Berlin: Language Science Press. DOI: [??](#).
- Welmers, William E. 1959. Tonemics, morphotonemics, and tonal morphemes. *General Linguistics* 5. 1–9.
- Welmers, William E. 1973. *African language structures*. Berkeley: University of California Press.
- Xu, Yi. 2001. Fundamental frequency peak delay in Mandarin. *Phonetica* 58(1-2). 26–52.
- Xu, Yi. 2004. Understanding tone from the perspective of production and perception. *Language and Linguistics* 5(4). 757–797.
- Xu, Yi & Xuejing Sun. 2002. Maximum speed of pitch change and how it may relate to speech. *The Journal of the Acoustical Society of America* 111(3). 1399–1413. DOI: [10.1121/1.1445789](https://doi.org/10.1121/1.1445789).

2 Defining tonal inventories: Criteria for toneme identification

Yip, Moira. 2002. *Tone* (Cambridge Textbooks in Linguistics). Cambridge University Press.